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Sensing Complete Injection for Animal Injection Device

Abstract

A semi-automated veterinary medicine delivery device may sense whether and how much air may have been incorporated into a dose delivered to an animal. The device may also sense an incomplete injection in some cases and may cause a second dose to be administered to make up for the incomplete dose. The device may compare a position sensor on a syringe in combination with a current sensor on a motor attached to the syringe to detect any abnormalities. In the case of an improper dose, the device may alert the user and administer a second dose.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This patent application claims priority to and benefit of U.S. patent application Ser. No. 16/863,423 filed 6 Oct. 2022 entitled “Sensing Complete Injection for Animal Injection Device,” and U.S. patent application Ser. No. 18/224,189 filed 20 Jul. 2023, the entire contents of which are hereby incorporated by reference for all they disclose and teach.

BACKGROUND

[0002] Vaccines, vitamins, medicines, and other substances are routinely administered to livestock. Some substances are injected into the animal intramuscularly, subcutaneously, intradermally, or intravenously. Other substances may be applied by drenching, oral administration, or other methods.

[0003] Animal husbandry is often performed by ranch hands in inhospitable environments. Ranch hands may have expertise in managing animals but may not be familiar with veterinary medicine delivery best practices. Further, many injections or medications may be applied in very rugged environments, subject to high heat or low temperatures, dust, dirt, rain, snow, and other weather.

SUMMARY

[0004] A semi-automated veterinary medicine delivery device may sense whether and how much air may have been incorporated into a dose delivered to an animal. The device may also sense an incomplete injection in some cases, and may cause a second dose to be administered to make up for the incomplete dose. The device may compare a position sensor on a syringe in combination with a current sensor on a motor attached to the syringe to detect any abnormalities. In the case of an improper dose, the device may alert the user and administer a second dose.

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In the drawings,

[0007] FIG. 1 is a diagram illustration of an embodiment showing a semi-automated dosing device in its ecosystem.

[0008] FIG. 2 is a diagram illustration of an embodiment showing a schematic or functional representation of a network with a semi-automated dosing device.

[0009] FIG. 3 is a flowchart illustration of an embodiment showing a method for dosing animals with re-dosing in case of an error.

[0010] FIG. 4 is a flowchart illustration of an embodiment showing a method for preparing a dosing device for use.

[0011] FIG. 5 is a flowchart illustration of an embodiment showing a method for dosing operations of a semi-automated dosing device.

[0012] FIG. 6 is a flowchart illustration of an embodiment showing a method for classifying errors during a dosing cycle.

[0013] FIG. 7 is a dimensionless chart showing energy signal vs distance traveled curves for different dosing situations.

[0014] FIG. 8 is a cutaway illustration of an embodiment showing a semi-automated dosing device.

[0015] FIG. 9 contains several illustrations of different adapters for a semi-automated dosing device.

DETAILED DESCRIPTION

Animal Injection Dosing System with Fault Detection

[0016] An animal injection system may detect abnormalities with its operation and may administer a second dose to make up for a deficiency identified in a first dose. An incomplete or partial dose may be sensed during the operation of an injection, and that portion of the dose that may have been improperly administered may be calculated and the device configured to administer a makeup dose.

[0017] An animal injection system may monitor an injection process to determine whether an injection is properly administered. Error conditions may be identified when air may be present in a dose, the dose may be administered in the wrong location or depth into an animal, an injection needle may be removed during the injection process, a needleless injection process may be improperly given when the device is moved sideways or lifted during the injection process, or other error conditions.

[0018] When an incomplete dose is detected, the device may alert a user and may configure the device to administer a second dose to the animal. In some cases, the entire dose may be re-administered, while in other cases, a partial dose may be re-administered. When a partial dose is given, the device may calculate a makeup dose based on the portions of the original dose that was improperly administered.

[0019] For a needle-based injection system, a device may compare a position sensor on a syringe with a current sensor on a motor used to drive the syringe. For a given medicine or other substance being delivered, an expected amount of current may be compared against the linear motion of the syringe. The comparison signal may be analyzed to identify different scenarios, such as air within the syringe, whether a needle used for injection was properly placed in the animal, whether the needle was removed prior to completing the injection, whether the syringe was properly filled prior to injection, and other conditions.

[0020] A semi-automated device may have a syringe barrel that may be filled by retracting a plunger to draw a substance into the barrel. Once loaded with the substance, a trigger may cause the plunger to be advanced and the substance will be expelled from the syringe. For injected substances, the device may be fitted with a needle or an adapter may be configured for needleless delivery. Other substances may be delivered by pour-on, nasally, or orally, where the device may be fitted with a tube or nozzle.

[0021] The syringe plunger may be driven by an electric motor, both in the retraction and extension stroke. During retraction, medicines or substances may be drawn into the syringe barrel through a feeder passage, and during extension, the substances may be expelled through a needle, tube, or other expulsion conduit. Both the feeder passage and expulsion conduit may be fitted with check valves. The check valve on the feeder passage may prevent substance in the syringe barrel from being forced out through the feeder passage during the extension stroke. Similarly, the check valve on the expulsion conduit may prevent air from entering the syringe barrel during the retraction stroke.

[0022] Some versions of the device may use a rack and pinion mechanism to extend and retract the

syringe plunger. The rack portion may be attached to the plunger, and the pinion may be operated by an electric motor. The rack portion may be attached to a position sensor, which may detect the linear motion travelled by the plunger. The electric motor may have a sensor to detect the motor's current draw. Other systems may use other mechanisms to drive a syringe and plunger, such as a chain and pulley mechanism, linear motors, or other mechanism.

[0023] A controller for the device may compare the linear motion of the syringe plunger with the current draw of the motor. By comparing the two signals, different conditions may be sensed during operation.

[0024] In normal operation, the device may retract the plunger to draw in the medicine or other substance into the syringe barrel. When a user positions the device ready for dosing, such as inserting the needle into the animal or arranging the oral drenching tube, the user may activate a trigger, which will cause the plunger to be extended to deliver the appropriate dose.

[0025] The motor may activate to deliver the dose and may advance the plunger forward until a measured dose is administered. In many cases, the syringe may be larger than a dose, and the motor may stop before fully emptying the syringe. In many cases, the weight or other characteristic of the animal may be used by the device controller to calculate an intended dose for that specific animal.

[0026] When a dose is smaller than the size of the syringe, the controller may calculate the size of the dose. In some cases, the dose from animal to animal may be identical, but in other systems, weight or other characteristics of the animal may vary the dose for each animal.

[0027] During the extension stroke when the substance is administered to an animal, the controller may monitor the linear motion of the plunger. The sensing of linear motion may have two uses: one may be to detect a predefined stop point to limit the dosage, and two may be to compare the linear motion to the current draw of the motor.

[0028] For normal operation, the viscosity of the medicine or other substance along with the orifice size of a needle or other delivery mechanism may determine how much force would be expected to be applied to the plunger to inject at a given rate. The force applied to the plunger may be directly proportional to the current draw of the motor per the distance travelled. When a dose is given within the normal range of force applied, the dose may be considered successfully administered.

[0029] When the measured force changes during delivery, a problem may be detected. When the force jumps from high resistance to low resistance, an air bubble may be assumed to be at the end of the chamber. In some cases, an air bubble may cause the force to be very low at the beginning then jump to the normal range. Such an air bubble may be at the exhaust or needle end of the syringe.

[0030] One common condition may be when a user removes an injection needle too soon during an injection. This may occur when a user inserts a needle into an animal, activates the trigger, and the injection process begins. During the injection process, an animal may jump or move, causing the needle to come out of the animal while the plunger is advancing. Sometimes, this situation may occur when the user inadvertently removes the injection needle without recognizing that the injection has not yet completed.

[0031] In such a situation, the device may sense that the force applied to the syringe may be in the normal range but then the force may change to be much less. At the point that the force changes, the controller may record the approximate dosage actually administered. The controller may alert the user that the dose has not finished and may calculate a second dose, where the second dose may be the approximate amount remaining from the first dose.

[0032] Other problematic situations may occur when a dose may be administered without an animal, where the dosage may be shot into the air or on the ground. Such a situation may occur when a user inadvertently presses a trigger to fire the dosing device.

[0033] Still another situation may occur when the dosage was administered in the wrong portion of the animal. Some medicines may be designed to be administered intramuscularly, while others may be intended to be delivered into fat, subcutaneously, or into some other portion of the animal's

body.

[0034] The resistance of the animal's tissue to an injection may be used as a signal to determine which layer of the animal's body received the injection. The signal may be compared to that of the expected layer of the animal's body and used to verify proper administration or to signal an error condition.

[0035] Some medicines or other substances may be administered in doses larger than the capacity of the syringe. In such cases, a complete dose may be administered in two, three, or more separate injections. Some medicines may be absorbed more readily when several smaller doses are administered rather than one larger dose.

[0036] In such a situation, the device may alert the user that there will be a set number of injections, and for example, the number may be three doses. The controller may indicate that the first dose is ready, when the user may place a needle in a first injection point and pull the trigger. After a successful injection, the device may re-load the syringe with more medicine, and the controller may indicate the second injection is ready. After inserting the needle in the second injection point and pulling the trigger, the device may administer the second dose. The third dose may follow the same procedure.

[0037] If one of the doses in a multi-dose sequence is detected to be incomplete, the controller may recalculate the next dose or doses to administer. In the case where an additional dose may be added because one or more of the previous doses in the multi-dose sequence was incomplete, the controller may notify the user and instruct the user to administer an additional dose.

[0038] With each dose, the controller may keep a record of the administered dose. In many cases, the identity of an animal being treated may be gathered, such as from an RFID tag, embedded chip, barcode, or other automated input. Some systems may have a manual input mechanism. A record of the actual dosage administered may be logged in a database. Such a database may be in the device controller or some other device.

[0039] For needleless injection devices, a sensor may be used to detect contact between the device and an animal's skin. The sensor may be a presence sensor or a force sensor, which may sense that the device remains in contact with the animal's skin during the injection process. Some systems may have a sensor that may detect lateral movement or sliding of the device across the surface of the animal's skin. During the injection sequence, a controller may monitor the sensor or sensors to determine whether proper contact and lack of movement was observed. If movement or lack of adequate pressure or contact were observed, the device may identify an improper injection and may configure itself to administer a makeup dose.

[0040] Throughout this specification and claims, the term “semi-automated dosing device” is used as a synonym for “dosing device” or, sometimes, “device” and refers to a device that applies doses to an animal with an inherent controller that can detect errors during dosing.

[0041] Throughout this specification, like reference numbers signify the same elements throughout the description of the figures.

[0042] In the specification and claims, references to “a processor” include multiple processors. In some cases, a process that may be performed by “a processor” may be actually performed by multiple processors on the same device or on different devices. For the purposes of this specification and claims, any reference to “a processor” shall include multiple processors, which may be on the same device or different devices, unless expressly specified otherwise.

[0043] When elements are referred to as being “connected” or “coupled,” the elements can be directly connected or coupled together or one or more intervening elements may also be present. In contrast, when elements are referred to as being “directly connected” or “directly coupled,” there are no intervening elements present.

[0044] The subject matter may be embodied as devices, systems, methods, and/or computer program products. Accordingly, some or all of the subject matter may be embodied in hardware and/or in software (including firmware, resident software, micro-code, state machines, gate arrays,

etc.) Furthermore, the subject matter may take the form of a computer program product on a computer-usable or computer-readable storage medium having computer-usable or computer-readable program code embodied in the medium for use by or in connection with an instruction execution system. In the context of this document, a computer-usable or computer-readable medium may be any medium that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

[0045] The computer-usable or computer-readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, device, or propagation medium. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media.

[0046] Computer storage media includes volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by an instruction execution system. Note that the computer-usable or computer-readable medium could be paper or another suitable medium upon which the program is printed, as the program can be electronically captured, via, for instance, optical scanning of the paper or other medium, then compiled, interpreted, or otherwise processed in a suitable manner, if necessary, and then stored in a computer memory.

[0047] When the subject matter is embodied in the general context of computer-executable instructions, the embodiment may comprise program modules, executed by one or more systems, computers, or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Typically, the functionality of the program modules may be combined or distributed as desired in various embodiments.

[0048] FIG. **1** is a diagram illustration of an example embodiment **100** showing an injection management system for livestock. Embodiment **100** may be one example of the various components that may administer and track the application of substances to livestock. In a typical use case, animals may be injected with vitamins, vaccines, or other medicinal substances. The system may validate that the doses were administered correctly and may keep a record of the dosing for various uses.

[0049] The system may identify whether or not a dose was properly administered at the point of administration. For example, a farmhand on a ranch may administer injections to livestock **102**. With each injection, a device **104** may verify that the dose was properly administered. When a dose was improperly administered, the device **104** may indicate to the operator, in this case, the farmhand, to re-administer a dose. The device **104** may configure itself for a makeup dose for the animal, and indicate to the operator to re-administer.

[0050] The device **104** may keep electronic records of each animal and dose. In many cases, the device **104** may use RFID, barcode, or other electronic communication to identify a specific animal or groups of animals, and add an electronic record of the dosing for the animals.

[0051] The device **104** may communicate with a wireless device **106**, which may transmit the records over a cellular network **108** to a computer network **110**.

[0052] In this example, a wireless device **106** may be a cellular telephone or other device that may be brought into the field. In some use cases, especially where cellular connections may not be available, the wireless device **106** may gather the electronic records from the device **104** and store the records. Upon connection to a cellular network **108** at a later time, the wireless device **106** may transmit the records.

[0053] The wireless device **106** may be used to program or configure the dosing device **104**. For

example, the dosing parameters for a group of animals may be configured using a user interface on the wireless device **106**. Once the dosing parameters are properly configured, the settings may be transmitted to the administration device **104**. In many systems, the administration device **104** may have a limited user interface but a wireless device **106** may have a larger, more easily accessible user interface. Such systems may be designed so that more complex configuration and monitoring tasks may be performed on the wireless device **106** rather than the administration device **104**.

[0054] The administration device **104** may have various user interface components. In some cases, the administration device **104** may have a digital display, where administrative and configuration functions may be performed, as well as monitoring and performance recording may be displayed.

[0055] The administration device **104** may have a simplified user interface that may be used during the routine of administering substances to animals. Such an interface may consist of a colored light or set of lights that may display different states of the device **104**. The states may include ready to administer, successful administration, unsuccessful administration, redosing previous animal, or other statuses of the device. In many cases, such a light or set of lights may be the primary user interface that an operator may use while actively administering medicines in the field.

[0056] The dosing data may be transmitted to a computer network **110**, where a management application **112** may manage and store configuration information in a configuration database **114** as well as store dosing information in an injection database **116**. The management application **112** may perform various administrative functions, such as veterinary functions, production monitoring, tracking and provenance monitoring, and other functions.

[0057] The use cases for the system of embodiment **100** may include immunizing herds or flocks against specific diseases. A dosing regimen may be designed by a veterinarian to immunize a herd by injecting 90% of the population. The device **104** may be configured to verify that each dose was properly administered and to record the doses. An immunization contractor may use the device **104** to administer the doses and record their doses, thereby verifying that the contractor has successfully completed their contract. A production manager may use the same information to insure their herd against a disease outbreak as the livestock matures. Finally, consumers may validate the provenance of their food through a record of an animal's history, which may include the dosing information.

[0058] The diagram of FIG. 2 illustrates functional components of a system. In some cases, the component may be a hardware component, a software component, or a combination of hardware and software. Some of the components may be application level software, while other components may be execution environment level components. In some cases, the connection of one component to another may be a close connection where two or more components are operating on a single hardware platform. In other cases, the connections may be made over network connections spanning long distances. Each embodiment may use different hardware, software, and interconnection architectures to achieve the functions described.

[0059] Embodiment **200** illustrates a device **202** that may have a hardware platform **204** and various software components. The device **202** as illustrated represents a conventional computing device, although other embodiments may have different configurations, architectures, or components.

[0060] The hardware platform **204** may include a processor **208**, random access memory **210**, and nonvolatile storage **212**. The hardware platform **204** may also include a user interface **214** and network interface **216**.

[0061] The random access memory **210** may be storage that contains data objects and executable code that can be quickly accessed by the processors **208**. In many embodiments, the random access memory **210** may have a high-speed bus connecting the memory **210** to the processors **208**.

[0062] The nonvolatile storage **212** may be storage that persists after the device **202** is shut down. The nonvolatile storage **212** may be any type of storage device, including hard disk, solid state memory devices, magnetic tape, optical storage, or other type of storage. The nonvolatile storage

212 may be read only or read/write capable. In some embodiments, the nonvolatile storage **212** may be cloud based, network storage, or other storage that may be accessed over a network connection.

[0063] The user interface **214** may be any type of hardware capable of displaying output and receiving input from a user. In many cases, the output display may be a graphical display monitor, although output devices may include lights and other visual output, audio output, kinetic actuator output, as well as other output devices. Conventional input devices may include keyboards and pointing devices such as a mouse, stylus, trackball, or other pointing device. Other input devices may include various sensors, including biometric input devices, audio and video input devices, and other sensors.

[0064] The network interface **216** may be any type of connection to another computer. In many embodiments, the network interface **216** may be a wired Ethernet connection. Other embodiments may include wired or wireless connections over various communication protocols.

[0065] The software components **206** may include an operating system **218** on which various software components and services may operate.

[0066] A dose controller **220** may be a routine that controls the administration of a dose by monitoring a current or energy sensor **222** and a linear position sensor **224** while a syringe plunger is advanced through a syringe body. The energy sensor **222** and linear position sensor **224** are one mechanism to measure force exerted while a syringe moves.

[0067] By monitoring the force exerted while a syringe plunger moves within a syringe body, the measured force may be compared to an expected force to determine whether a dose was administered as expected or whether an error may have occurred. Further, the type of error may be identified by analysis of the force signal.

[0068] One error that may occur is that air may have been captured in the syringe body. In general, air is highly compressible while liquid doses may not be. When air is present, the force applied to a syringe body may be much different than when a syringe body contains no air. This difference can be identified to detect that air is present.

[0069] Further, the amount of air present may be calculated by signal analysis. When the amount of air can be calculated, a make up dose may be calculated. For example, a dose that may be administered with a syringe that contains 60% medicine and 40% air may have a second, makeup dose calculated to be 40% of the original dose. When the dose controller **220** detects such an incomplete dose, an alert may be made to an operator, who may then administer the makeup dose to the animal.

[0070] Another type of error may occur when a dose may be administered without an animal present. Such a situation may occur when a trigger **226** may be activated prior to inserting an injection needle into an animal. When the device **202** activates the syringe and no animal is present, the force detected by a current or energy sensor **222** and a linear position sensor **224** may give a much different signal than expected. Such a signal may also be different from a signal when air is present in the syringe, and thereby the device **202** may be capable of isolating such an error.

[0071] Yet another type of error may occur when an injection location was improper. Some medicines are intended to be delivered intramuscularly, while others are to be delivered subcutaneously, in the animal's fatty tissue, or some other location. Each of these various locations may provide a different level of resistance during injection, and by analyzing the force signal during injection, the dose controller **220** may be able to differentiate between the intended and an improper location for the dose.

[0072] In some cases, an error may occur because the position of an injection needle may change during dosing. An animal may jerk or move, or an operator may withdraw the needle while an injection occurs. In such a case, an injection may begin properly, but may change during the injection sequence. By monitoring the force signal, such changes may be identified and an error condition may be isolated.

[0073] Such an error condition may be identified and communicated to the device operator. The communication may alert the operator that the injection was not in the right location or that the injection was incomplete. Such feedback may help the operator correctly place the needle for future doses.

[0074] In some such situations, the dose controller **220** may determine that redosing may be appropriate, while in other situations, the improper dose may be logged but may not be sufficiently bad to warrant redosing.

[0075] The force exerted on a syringe plunger within a syringe body may be monitored in many different manners. In embodiment **200**, a current or energy sensor **222** may be coupled with a linear position sensor **224**. A current or energy sensor **222** may measure current draw to an electric motor, while the linear position sensor **224** may measure physical movement. The combination of the two measurements may yield force applied to the syringe.

[0076] Other mechanisms may be used to generate similar signals, including force sensors placed on the syringe mechanism, pressure sensors within the syringe body, or other sensors and locations. Such sensors may generate similar signals that may be generated using the energy sensor **222** and linear position sensor **224**.

[0077] The dose controller **220** may communicate with a human operator using a farm hand interface **228**. The farm hand interface **228** may be the interface through which an operator may be given the status of the device **202**, as well as alerts to administer doses, re-administer makeup doses, success or failure of doses, and other status or alerts. In many cases, the farm hand interface **228** may not allow the operator to make changes to the dosing or perform other administrative changes to the device. A typical user interface may be as simple as a single light, a set of lights, multicolored lights, or some other simple signal.

[0078] Some devices **202** may include a graphical, text, or other user interface. Such interfaces may display a count of doses administered, text based or graphical representations of the success or failure of the doses, and other information.

[0079] With each operation of the device **202**, records may be written to the dosage records **230**. The dosage records **230** may be a local repository of doses. In many use cases, the device **202** may store dose information, then may transmit that information to another device at some later point. In many use scenarios, doses may be administered on a ranch or some other rural location that may be many miles from a connection to a cellular or other communications network.

[0080] An administrative or dosage interface **232** may be a mechanism by which the device **202** may be configured and managed. Through such an interface, the device **202** may be configured, updated, and managed. In many cases, the interface **232** may have the capability of configuring the dosage information to be applied to animals. In some embodiments, such functions may be available to managers or veterinarians, and in such embodiments, the farm hand operators may not be able to adjust dosage or reconfigure the device **202**.

[0081] The device **202** may have various adapters **234**. Different adapters **234** may have needle injection devices, needleless injection, intranasal delivery, oral or pour-on delivery, or other methods of medicine delivery. Such designs may allow a standardized “gun” device **202** to be reconfigured for different use cases. Each adapter **234** may have an electrical identifier **236** as well as a mechanical interface **238**.

[0082] An electrical identifier **236** may be an RFID, barcode, chip, or other electrically-readable identifier that the device **202** may query to determine which adapter **234** has been installed. In many cases, a dosage may be defined to be applied in a specific manner, and such a dosage definition may include an adapter definition for that dosage. During normal operation, the dose controller **220** may compare the required adapter definition for the dose with the adapter's electrical identifier **236** to validate that the proper adapter is present.

[0083] The mechanical interface **238** may be a mechanical connection between the device **202** and the adapter **234**. The mechanical interface **238** may vary from one design to the next, and may

include the applicator portion that may come in contact with an animal. In an example of needle-based injections, the adapter **234** may be quickly replaceable so that needles may be replaced after so many uses.

[0084] A medicine interface **240** may include an RFID or barcode identifier **242**. The medicine interface **240** may be one way that the medicine may be logged by the dose controller **220** in the dosage records **230**. In many cases, a medicine container may have radio frequency identifiers (RFID), barcode, or other machine-readable identifier. The device **202** may capture the identifier and compare the actual identifier with the medicine identifier intended for the dosing to be administered.

[0085] Some systems may operate with a scale **246** or other animal interfaces **248**. Such interfaces may include temperature measurements **250**, visual capture devices **252**, animal identifiers **254**, or other inputs. Parameters collected by these devices may be used to calculate dosing as well as record keeping for the dosing.

[0086] Inputs such as animal weight gathered from a scale **246** or animal temperature **250** may be used to calculate dosing for a specific animal. In many cases, dosing may be defined as a function of weight, age, temperature, or other factor. The dose controller **220** may be programmed to calculate a dose using such factors for each animal, and a record of the weight or other factor may be stored with the dosage in the dosage records **230**.

[0087] The device **202** may be connected to a network **256** to various other devices.

[0088] One such device may be a local administrative device **258**. The local administrative device **258** may operate on a hardware platform **260** and may have an administrative or maintenance interface **262** as well as a dosage controller **264**.

[0089] In many use cases, the local administrative device **258** may be used to program and manage the device **202**, and may store dosage records **266** gathered from the device's dosage records **230** for later communication to other computer systems. In many systems, a single local administrative device **258** may be used to manage several devices **202** in the field.

[0090] The local administrative device **258** may have a dosage controller **264** which may operate with a configuration database **268** to create and manage doses. The configuration database **268** may have configuration information and options available for the device **202**, including typical dosing information for specific medicines and other information. Such a database may be used by someone in the field to configure the device **202**.

[0091] A cloud services manager **270** may be a service operable in a cloud architecture that may provide management, record keeping, and configuration services for the device **202** and local administrative device **258**.

[0092] A database of dosage records **272** may be gathered from the device **202** or local administrative device **258**. Such a database may be accessed by a veterinarian interface **278**, a production manager interface **274**, or other interface.

[0093] A production manager interface **274** may be used by a supervisor of a livestock operation to track the animals under their care. The production manager interface **274** may allow such a supervisor to schedule doses, monitor when the doses were administered, and capture other information regarding the livestock.

[0094] A veterinarian interface **278** may be used by a veterinarian to review dosage administration and other parameters regarding the health and well being of the livestock.

[0095] Both the production manager interface **274** and veterinarian interface **278** may interact with a device configuration database **276** to create dosage regimes for livestock. The dosage information and device configurations may be set and transmitted to the local administrative device **258** or device **202**. Once transmitted to the device **202**, a farm hand may administer the doses to the animals.

[0096] FIG. **3** is a flowchart illustration of an embodiment **300** showing a method for dosing animals with a re-dosing process for unsuccessful doses. The operations of embodiment **300** show

one workflow of how a semi-automated dosage system might be deployed to administer medicines, vaccines, or other substances to livestock.

[0097] Other embodiments may use different sequencing, additional or fewer steps, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations or set of operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The steps selected here were chosen to illustrate some principles of operations in a simplified form.

[0098] A device may be configured in block **302**. The configuration may include receiving parameters defining how a dose will be administered, such as dosage size, the configuration of the dosing device, adapters to be used, any formulas used to calculate dosing, and the like. The configuration may also include an expected force-related signal that the application device may experience during normal operation.

[0099] The force-related signal may be defined in a manner such that a dosing device may be able to determine whether a dose was properly administered as well as any error conditions. One such error condition may be to calculate how much of a dose was properly administered such that a makeup dose may be calculated and administered.

[0100] An animal may be identified in block **304**. In a typical use case, an animal's RFID tag may be read, for example. The animal's weight or other parameters may be gathered in block **306**.

[0101] A dosage may be calculated in block **308** and the device may be configured for the dose in block **308**. The dose may be applied in block **310**.

[0102] If the dose was a success in block **312**, the record of the dose may be stored in block **314**, and if another animal is to be dosed in block **316**, the process may return to block **304**.

[0103] If the dose was not a success in block **312**, the improper dose may be recorded in block **318** and a determination may be made whether a follow up or makeup dose may be made in block **320**.

If a makeup dose is to be performed in block **320**, the new dose may be calculated in block **322**. The operator may be notified in block **324**, and the process may return to block **308** to configure and administer the makeup dose. The process may loop back again if the second dose was not proper as well.

[0104] After all animals have been processed in block **316**, a waiting loop may occur in block **326** until a connection is made with a network. Once the network connection is established in block **326**, the dosing records may be uploaded in block **328**.

[0105] FIG. **4** is a flowchart illustration of an embodiment **400** showing a method for preparing a semi-automated dosing device for use. The operations of embodiment **300** show one workflow for deploying a dosing regimen, configuring a device, and going through a priming sequence. The workflow of embodiment **400** illustrate merely one sequence that may be deployed to get a device operational for dosing.

[0106] Other embodiments may use different sequencing, additional or fewer steps, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations or set of operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The steps selected here were chosen to illustrate some principles of operations in a simplified form.

[0107] A set of functions may be performed by an administrative interface in block **402**. The administrative functions may identify parameters for treating animals in block **404**, creating a log table for the dosing records in block **406**, and uploading the configuration parameters to the dosing device in block **408**.

[0108] The parameters for treating animals in block **404** may include identifying the animals being treated. In many cases, the animals may be identified by species, herd location, and, when known, the individual animals for which treatment is prescribed.

[0109] The parameters may also include information about the medicine or other substance to be administered. The substance information may include the identifiers for the medicine as well as

physical characteristics. One characteristic may be viscosity of the substance, and in some cases, a viscosity curve that gives viscosity as a function of temperature. Viscosity may affect the amount of force expected to be applied to a syringe to draw the substance into the syringe body and to expel the substance.

[0110] The parameters may include conditions under which the medicines may be administered. The conditions may include the environmental conditions, such as outdoor temperature range, but may also include the location within the animal where the substance may be delivered. Such a location may be intramuscular, into fatty tissue, subcutaneous, intranasal, orally, drenching on the surface of the animal, or some other location. Such location information may inform the expected resistance level that may be experienced by a force measurement taken during the injection process.

[0111] A device configuration block **410** may include the mechanical and electrical configuration of the device. An adapter may be mechanically attached in block **412** and scanned electrically in block **414** to determine the adapter's identification. If the wrong adapter is attached in block **416**, an error may be flagged in block **418** and the operator alerted.

[0112] Providing that the adapter is correct in block **416**, the medicine may be connected to the device in block **420** and scanned in block **422**. Different devices may have different types of reservoirs of medicine or other substance to be injected. In some cases, a reservoir may be worn on an armband, slung around one or both shoulders, or otherwise connected to an application device through a hose or tube. If the medicine is incorrect in block **424**, an error may be flagged in block **426** and an operator alerted. Once the appropriate adapter and medicine has been configured, the device may be ready for priming.

[0113] The priming sequence may be in block **428**. The sequence may begin in block **430**, and the operator may be alerted in block **432**. The operator may activate a trigger in block **434**, where the plunger may eject whatever is in the syringe body in block **436**, then pull in a dose into the syringe body or chamber in block **438**.

[0114] While pulling in the substance to the syringe body, a dose controller may monitor the force applied to the syringe plunger. The force may be measured, for example, through an energy or current sensor on an electric motor along with the linear motion measured for the syringe plunger's movement. When the force deviates from the expected force profile of a full chamber in block **440**, the operator may be alerted in block **446**, and the operator may again trigger the operation in block **434**.

[0115] A priming operation may pump the syringe several times to load the syringe body with a full compliment of medicine. When initially configured, air may be present in the chamber, tubing, and other components. As the priming operation continues, the air may be removed and the lines and chamber filled with the substance to be injected. Once the system has been properly primed, an alert may be made to indicate to the operator that dosing may begin in block **448**.

[0116] FIG. 5 is a flowchart illustration of an embodiment **500** showing a method for dosing animals. The operations of embodiment **500** show operations performed by an operator **502** and a semi-automated dosing device **504**.

[0117] Other embodiments may use different sequencing, additional or fewer steps, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations or set of operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The steps selected here were chosen to illustrate some principles of operations in a simplified form.

[0118] The device **504** may alert an operator to begin dosing in block **506**. The operator **502** may receive the alert in block **508** and may position the device for dosing on the animal in block **510**, then activate the device trigger in block **512**.

[0119] The positioning operation of block **510** may be different for various ways of applying the substance to an animal. In a needle injection process, a needle may be inserted into the animal's

body in the appropriate location, depth, and angle for the intended delivery. In a needleless injection, the needleless injection adapter may be pressed against the animal's body with the appropriate force. In an intranasal delivery, the application device may be inserted into an animal's nasal passage. In an oral delivery, the application device may be inserted into an animal's throat. In a pour on delivery, the application device may be placed on the animal's back to deliver to the animal's skin.

[0120] A trigger signal may be received in block **514** by the device **504**. The animal identification may be scanned in block **516** and any parameters associated with the animal may be retrieved in block **518**. The parameters may include, for example, the animal's weight.

[0121] The dosage may be calculated in block **520** and an expected force profile may be determined in block **522**.

[0122] The syringe plunger may begin advancing in block **524** to deliver the substance to the animal. As the plunger advances, linear movement may be gathered from a linear movement sensor in block **526** and energy or current consumed may be gathered in block **528**. Such measurements may occur in real time during the motion of the plunger.

[0123] If the measurements are not staying within an expected profile, the portion of the movement that is not OK is recorded in block **532**. If the dosage has not completed, the process may return to block **526** to continue monitoring as the movement occurs. The monitoring and measurement may occur continually during the movement of the syringe plunger.

[0124] Once the syringe plunger has moved according to the dose in block **534**, the dose record may be stored. If the dose was not a success in block **538**, a make up dose may be calculated in block **540**. The operator may be alerted in block **542**, and the process may return to block **514** to receive another trigger signal and administer a makeup dose.

[0125] If the dose was a success in block **538**, an alert may be given in block **544** that the dose was successful. If the operator **502** has additional animals to dose in block **546**, the process may return to block **508**. If not, the process may terminate in block **548**.

[0126] FIG. **6** is a flowchart illustration of an embodiment **600** showing a method for classifying error during dose administration using a semi-automated dosing device. The operations of embodiment **600** show one workflow of how a semi-automated dosage system might identify specific error conditions that may occur during dosing.

[0127] Other embodiments may use different sequencing, additional or fewer steps, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations or set of operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The steps selected here were chosen to illustrate some principles of operations in a simplified form.

[0128] Embodiment **600** illustrates several different error conditions that may be detected by analyzing force applied to a syringe plunger. These error conditions may be used to correct a detected problem by re-administering a dose.

[0129] Embodiment **600** begins at the stage where a syringe plunger is being advanced during a dosing event in block **602**. During the movement, the linear movement and energy consumption may be monitored in block **604**. In embodiments where a force sensor, internal pressure sensor, or another sensor that measures forces applied directly, such a sensor may be monitored in lieu of a separate linear motion sensor and current or energy consumption sensor.

[0130] If the forces being exerted are meeting the expected profile in block **606**, and the dose has not completed in block **608**, the process loops back to block **604**.

[0131] If the forces being exerted are not meeting the expected profile in block **606**, several different types of errors may be detected. In block **610**, an empty dose may be detected. An empty dose may be a dose where the syringe may be filled with air and provides very little resistance while expelling the dose. Such a situation may be logged as an empty dose in block **612**. In many cases, an empty dose may cause a priming operation to be performed, followed by a second attempt

at dosing the animal.

[0132] A dose may be made with a full chamber of a syringe, yet the animal may not be present in block **614**. Such a situation may happen when an operator presses the trigger to begin operation prior to positioning the device appropriately. In a needle-based injection, the needle may be inserted into the animal's body. In a needleless injection system, the injection point may be placed against the animal's body. In an intranasal delivery, the applicator may be placed in the animal's nose.

[0133] For some pour on or oral delivery systems, the animal may not create any detectable resistance, so such an error condition may not be sensed by a force sensor as described.

[0134] When a dose may be given in block **614** when no animal is present, the dose may be logged as no animal detected in block **616**. In many cases, such an error may cause the device to configure itself to re-apply the same dose and may alert the operator to re-apply the dose to the same animal.

[0135] In some situations, the forces being exerted may indicate that a partial dose was applied. A partial dose may occur when an operator removes a needle or other applicator during the dosing process, which may happen, for example, when an animal moves during the process. Another cause of a partial dose may occur when the syringe may contain some air and some medicine. In such a case, a partial dose of medicine may be delivered.

[0136] When a dose was detected to be partial, the dose may be logged as such in block **620**. A calculation may be performed in block **622** to determine how much of the dose was properly administered and how much was not. Such a calculation may be used to determine the size of a makeup dose.

[0137] A dose may be placed in an improper location in block **624**. An improper location may be in the wrong portion of an animal. For example, a subcutaneous injection may be improperly placed in the animal's muscle or fat tissue. Another example may be an intramuscular injection that may be placed too shallow, such as in fatty tissue or subcutaneously.

[0138] An improper location may be detected when the resistance of particular portions of the animal's anatomy may be different enough that an analysis of force curves may identify improper location. When a dose may be identified as improperly located in block **624**, the dose may be logged in block **626** and the proper or intended location may be identified in block **628**. An alert may be sent to an operator so that the operator may correct future doses. In some situations, a dose may be re-administered.

[0139] If a deviation from the force profile may be detected in block **606** but not otherwise classified, the dose may be logged as having an unknown or other type of error in block **630**.

[0140] Once an error has been detected, the process may return to block **608** and may continually monitor the remaining portion of the dose administration.

[0141] When the dose has completed in block **608**, if an error has not occurred in block **632**, a successful dose may be identified in block **636**. If an error has occurred in block **632**, the operator may be alerted in block **634** and, in some cases, the dose may be re-administered in block **638**.

[0142] FIG. 7 is an illustration of an embodiment **700** showing several force curves. In embodiment **700**, a dimensionless signal **702** is shown on the Y axis and dimensionless position is shown in the X axis. The curves illustrate measurements of current or energy consumed as a syringe plunger is driven forward to expel a medicine from an injection-type semi-automated dose administration device.

[0143] The curve **706** may be a successful dose, which the curve **708** may be a partial dose, where 40% of the chamber may have been filled with air. The successful dose shows a high rate of current draw initially, then continues at a high level during the dose administration. The curve showing a partially filled chamber is similar but offset. At the beginning of the partially filled curve **708**, the initial force is lower than the fully filled dose curve **706**, but once the air is compressed at the first measurement, the force curve **708** follows about the same shape as the fully filled curve **706**.

[0144] The curves **710** and **712** show no animal detected in curve **710** and an instance of an empty

chamber in curve **712**. The no animal detected curve **710** may occur with a fully filled chamber, but since the animal may not be present and offers no resistance, the signal measurement drops initially. The signal measurement increases over time, as the resistance of the medicine flowing through the needle increases.

[0145] In contrast, the curve **712** shows an empty chamber. This curve starts out similar to the no animal detected curve **710**, but continues with very little resistance because the lack of a viscous fluid to provide resistance.

[0146] The curves of embodiment **700** illustrate that analysis of force curves or current/energy vs linear movement may be used to isolate different error conditions that may occur during use of a semi-automated dosing device. When an unsuccessful dose may be identified, the dose may be logged and an appropriate remediation may be applied. In some cases, the entire dose may be re-administered. In cases where the dose may be partially administered, a makeup dose may be calculated and re-administered.

[0147] FIG. **8** is a cut-away illustration of an embodiment **800** showing a semi-automated dosing device.

[0148] The device **800** has a frame **802**, power source **804**, and a DC motor **806**.

[0149] The DC motor **806** may drive a gearbox **810** and a belt **812**, which is part of a linear drivetrain **814** which drives the syringe plunger **822**. A linear position sensor **816** and energy or current sensor **818** may combine to determine the force being applied to the plunger **822** as the plunger **822** travels through the fluid chamber **820**.

[0150] An adapter **824** may be illustrated as a needle-based injection system. Medicine or other substance may be introduced through an inlet **826**. When the plunger **822** is moved to the right (as illustrated), a vacuum may be created inside the fluid chamber **820**, drawing the substance past a check valve **828** and filling the fluid chamber **820**, also known as a syringe body. During the fill process, a check valve **830** may be positioned so that air is not drawn into the fluid chamber **820** from the needle or administration end of the device.

[0151] When the plunger **822** is moved to the left (as illustrated), the check valve **828** from the inlet feed may be pushed closed and the check valve **830** at the administration end may open, allowing the substance to exit the device through the outlet **832**.

[0152] The device **800** may be outfitted with a controller **836**, which may be in electrical contact with the various sensors as well as a user display **838** mounted on the top of the device. Further, a status indicator **840** may be a light, multicolored light, or set of lights that may give a quick visual status of the device during operation.

[0153] FIG. **9** is a cutaway illustration of several different adapters that may be attached to the device of embodiment **800**.

[0154] An injection adapter **902** may have a needle **904** placed behind a retractable shroud **906**. The adapter **902** may have a check valve **908** illustrated.

[0155] A needleless injection adapter **910** may be illustrated with a nozzle **912**, which may inject a substance through a high velocity flow of medicine. A check valve **914** is also illustrated.

[0156] An intranasal adapter **916** may show a nozzle **918** designed for applying medicines or other substances into an animal's nose. A check valve **920** is shown.

[0157] An oral or pour-on adapter **922** is illustrated with a nozzle **924** and a check valve **926**.

[0158] The foregoing description of the subject matter has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the subject matter to the precise form disclosed, and other modifications and variations may be possible in light of the above teachings. The embodiment was chosen and described in order to best explain the principles of the invention and its practical application to thereby enable others skilled in the art to best utilize the invention in various embodiments and various modifications as are suited to the particular use contemplated. It is intended that the appended claims be construed to include other alternative embodiments except insofar as limited by the prior art.

Claims

1. A method for managing dose application for animals, said method comprising: receiving an animal identifier for a first animal; determining a first set of parameters for said first animal; determining a first dose for said first animal based on said first set of parameters; configuring a first dosing device to inject said first animal using said first dose; determining that said first dosing device was not successful in applying said first dose and storing a first parameter associated with said first dose and said first animal; determining a second dose for said first animal, said second dose being determined at least in part from said at least one parameter associated with said first dose; configuring a second dosing device to inject said first animal using said second dose; determining that said second dosing device was successful in applying said second dose and storing a second parameter associated with said second dose and said first animal.
 2. The method of claim 1, said first dosing device being the same as said second dosing device.
 3. The method of claim 2 further comprising: sending a first notification to a user of said first dosing device that said first dose was not successful.
 4. The method of claim 3 further comprising: sending a second notification to said user that said second dosing device is configured to apply said second dose.
 5. The method of claim 1 further comprising: receiving an animal identifier for a second animal; determining a second set of parameters for said second animal; determining a third dose for said second animal based on said second set of parameters; configuring a first dosing device to inject said second animal using said third dose; determining that said first dosing device was successful in applying said third dose and storing a third parameter associated with said third dose and said second animal.
 6. The method of claim 5 being performed in its entirety by a controller of said first dosing device.
 7. The method of claim 6, said first set of parameters comprising a weight for said first animal.
 8. The method of claim 7, said weight being electronically transmitted to said controller.
 9. The method of claim 8 further comprising: receiving a product identifier for said first dose.
 10. The method of claim 6 further comprising: receiving a trigger input, said trigger input causing said first dosing device to administer said first dose.
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